

# Assignment\_4

**1.A Developer is running VM but required additional storage for its data create a volume of required size; attach the volume to existing instance; verify the volume inside vm; create a file system; mount a volume and store test file in it;**

In that go to the Login Page :

## Step 1:

Create An Instance in that,

The screenshot shows the OpenStack dashboard interface. The left sidebar contains navigation links for Project, API Access, Compute, Overview, Instances, Images, Key Pairs, Server Groups, Volumes, Network, Admin, and Identity. The main content area is titled 'Instances' and shows a table with one instance. The instance is named 'hpcal' and is in a 'Running' state. The table columns are: Instance Name, Image Name, IP Address, Flavor, Key Pair, Status, Availability Zone, Task, Power State, Age, and Actions. The instance 'hpcal' is using the 'cirros-0.6.3-x86\_64-disk' image, has an IP address of '172.24.4.165, 2001:db8::302', and is running on the 'm1.nano' flavor. The 'Actions' column for this instance includes a 'Create Snapshot' button.

| Instance Name | Image Name               | IP Address                  | Flavor  | Key Pair | Status | Availability Zone | Task | Power State | Age       | Actions         |
|---------------|--------------------------|-----------------------------|---------|----------|--------|-------------------|------|-------------|-----------|-----------------|
| hpcal         | cirros-0.6.3-x86_64-disk | 172.24.4.165, 2001:db8::302 | m1.nano | -        | Active | nova              | None | Running     | 0 minutes | Create Snapshot |

Launch the Instance,

Launch Instance

Details \*

Source \*

Flavor \*

Networks \*

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Please provide the initial hostname for the instance, the availability zone where it will be deployed, and the instance count. Increase the Count to create multiple instances with the same settings.

Project Name

admin

Instance Name \*

Description

Availability Zone

nova

Count \*

1

Total Instances  
(10 Max)

20%

1 Current Usage

1 Added

8 Remaining

× Cancel

< Back

Next >

Launch Instance

Instance Info,

## Launch Instance



Details

Source

Flavor \*

Networks \*

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Instance source is the template used to create an instance. You can use an image, a snapshot of an instance (image snapshot), a volume or a volume snapshot (if enabled). You can also choose to use persistent storage by creating a new volume.

### Select Boot Source

Image

### Create New Volume

Yes No

### Volume Size (GB) \*

1

### Delete Volume on Instance Delete

Yes No

### Volume Type

lvmdriver-1

### Allocated

Displaying 1 item

| Name                       | Updated          | Size     | Format | Visibility |   |
|----------------------------|------------------|----------|--------|------------|---|
| > cirros-0.6.3-x86_64-disk | 9/14/26 12:16 PM | 20.69 MB | QCOW2  | Public     | ↓ |

Displaying 1 item

### ▼ Available 0

Select one

Click here for filters or full text search.

Displaying 0 items

| Name                 | Updated | Size | Format | Visibility |
|----------------------|---------|------|--------|------------|
| No items to display. |         |      |        |            |

Displaying 0 items

× Cancel

< Back

Next >

Launch Instance

## Launch Instance



Details

Source

Flavor

Networks \*

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Flavors manage the sizing for the compute, memory and storage capacity of the instance.

### Allocated

Displaying 1 item

|   | Name    | vCPUs | RAM    | Total Disk | Root Disk | Ephemeral Disk | Public |   |
|---|---------|-------|--------|------------|-----------|----------------|--------|---|
| > | m1.nano | 1     | 192 MB | 1 GB       | 1 GB      | 0 GB           | Yes    | ↓ |

Displaying 1 item

### Available 11

Select one

Displaying 11 items

|   | Name      | vCPUs | RAM    | Total Disk | Root Disk | Ephemeral Disk | Public |   |
|---|-----------|-------|--------|------------|-----------|----------------|--------|---|
| > | m1.micro  | 1     | 256 MB | 1 GB       | 1 GB      | 0 GB           | Yes    | ↑ |
| > | cirros256 | 1     | 256 MB | 1 GB       | 1 GB      | 0 GB           | Yes    | ↑ |
| > | m1.tiny   | 1     | 512 MB | 1 GB       | 1 GB      | 0 GB           | Yes    | ↑ |
| > | ds512M    | 1     | 512 MB | 5 GB       | 5 GB      | 0 GB           | Yes    | ↑ |
| > | ds1G      | 1     | 1 GB   | 10 GB      | 10 GB     | 0 GB           | Yes    | ↑ |
| > | m1.small  | 1     | 2 GB   | 20 GB      | 20 GB     | 0 GB           | Yes    | ↑ |
| > | ds2G      | 2     | 2 GB   | 10 GB      | 10 GB     | 0 GB           | Yes    | ↑ |
| > | m1.medium | 2     | 4 GB   | 40 GB      | 40 GB     | 0 GB           | Yes    | ↑ |
| > | ds4G      | 4     | 4 GB   | 20 GB      | 20 GB     | 0 GB           | Yes    | ↑ |
| > | m1.large  | 4     | 8 GB   | 80 GB      | 80 GB     | 0 GB           | Yes    | ↑ |



Launch Instance

Details

Source

Flavor

**Networks**

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Networks provide the communication channels for instances in the cloud. You can select ports instead of networks or a mix of both.

▼ Allocated <sup>1</sup>

Displaying 1 item

| Network  | Subnets Associated                  | Shared | Admin State | Status |
|----------|-------------------------------------|--------|-------------|--------|
| > public | ipv6-public-subnet<br>public-subnet | No     | Up          | Active |

Displaying 1 item

▼ Available <sup>1</sup>

Select one or more

Click here for filters or full text search.

Displaying 1 item

| Network  | Subnets Associated | Shared | Admin State | Status |
|----------|--------------------|--------|-------------|--------|
| > shared | shared-subnet      | No     | Up          | Active |

Displaying 1 item

Cancel

< Back

Next >

Launch Instance

Then Instance is Created,

openstack

admin

Project

API Access

Compute

Overview

**Instances**

Images

Key Pairs

Server Groups

Volumes

Network

Admin

Identity

Project / Compute / Instances

Instances

Instance ID

Filter

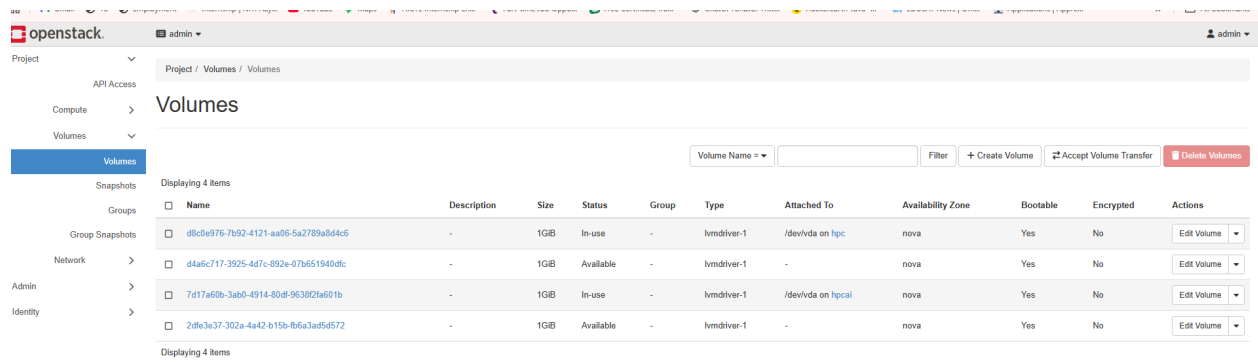
Launch Instance

Delete Instances

More Actions

| Instance Name | Image Name               | IP Address                  | Flavor  | Key Pair | Status | Availability Zone | Task | Power State | Age       | Actions         |
|---------------|--------------------------|-----------------------------|---------|----------|--------|-------------------|------|-------------|-----------|-----------------|
| hpc           | -                        | 172.24.4.142, 2001:db8::324 | m1.nano | -        | Active | nova              | None | Running     | 0 minutes | Create Snapshot |
| hpcal         | cirros-0.6.3-x86_64-disk | 172.24.4.165, 2001:db8::302 | m1.nano | -        | Active | nova              | None | Running     | 5 minutes | Create Snapshot |

## Then Go for the Volume Add the Volume



The screenshot shows the OpenStack Volumes dashboard. The left sidebar contains navigation links for Project, API Access, Compute, Volumes, Snapshots, Groups, Group Snapshots, Network, Admin, and Identity. The main content area is titled 'Volumes' and displays a table of four volumes. Above the table are controls for filtering and creating volumes.

Project / Volumes / Volumes

Volumes

Volume Name  Filter + Create Volume Accept Volume Transfer Delete Volumes

Displaying 4 Items

| <input type="checkbox"/> | Name                                 | Description | Size | Status    | Group | Type        | Attached To       | Availability Zone | Bootable | Encrypted | Actions                     |
|--------------------------|--------------------------------------|-------------|------|-----------|-------|-------------|-------------------|-------------------|----------|-----------|-----------------------------|
| <input type="checkbox"/> | d8c0e976-7b52-4121-aa06-5a2785a8d4c6 | -           | 1GB  | In-use    | -     | lvmdriver-1 | /dev/vda on lpc   | nova              | Yes      | No        | <a href="#">Edit Volume</a> |
| <input type="checkbox"/> | d4a6c717-3925-4d7c-892e-87b651940dfc | -           | 1GB  | Available | -     | lvmdriver-1 | -                 | nova              | Yes      | No        | <a href="#">Edit Volume</a> |
| <input type="checkbox"/> | 7d17a60b-3ab0-4914-80df-563802f601b  | -           | 1GB  | In-use    | -     | lvmdriver-1 | /dev/vda on hpcal | nova              | Yes      | No        | <a href="#">Edit Volume</a> |
| <input type="checkbox"/> | 2df63a37-302a-4a42-b15b-8b6a3ad5d572 | -           | 1GB  | Available | -     | lvmdriver-1 | -                 | nova              | Yes      | No        | <a href="#">Edit Volume</a> |

Displaying 4 Items

## Create the Volume of 5 GB

Create Volume

Volume Name

hpcvol

Description

Volume Source

No source, empty volume

Type

lvmdriver-1

Size (GiB) \*

5

Availability Zone

nova

Group

No group

Description:

Volumes are block devices that can be attached to instances.

Volume Type Description:

lvmdriver-1

No description available.

Volume Limits

Total Gibibytes

4 of 1,000 GiB Used

Number of Volumes

4 of 10 Used

Cancel

Create Volume

Attach the Volume to the instance:

Project / Compute / Instances

Instances

Instance ID

Filter

Launch Instance

Delete Instances

More Actions

Displaying 2 Items

| <input type="checkbox"/> | Instance Name | Image Name               | IP Address                  | Flavor  | Key Pair | Status | Availability Zone | Task | Power State | Age       | Actions                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------|---------------|--------------------------|-----------------------------|---------|----------|--------|-------------------|------|-------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | hpc           | cirros-0.6.3-x86_64-disk | 172.24.4.142, 2001:db8::324 | m1.nano | -        | Active | nova              | None | Running     | 2 minutes | Create Snapshot                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <input type="checkbox"/> | hpcal         | cirros-0.6.3-x86_64-disk | 172.24.4.165, 2001:db8::302 | m1.nano | -        | Active | nova              | None | Running     | 8 minutes | Associate Floating IP<br>Attach Interface<br>Detach Interface<br>Edit Instance<br>Attach Volume<br>Detach Volume<br>Update Metadata<br>Edit Security Groups<br>Edit Port Security Groups<br>Console<br>View Log<br>Rescue Instance<br>Pause Instance<br>Suspend Instance<br>Shelve Instance<br>Resize Instance<br>Lock Instance<br>Soft Reboot Instance<br>Hard Reboot Instance<br>Migrate Instance<br>Shut Off Instance<br>Rebuild Instance |

Displaying 2 Items

Attach the Vol,

admin

Project / Compute / Instances

Instances

Volume ID

Description:

Attach Volume to Running Instance.

Cancel

Attach Volume

Displaying 2 Items

| <input type="checkbox"/> | Instance Name | Image Name               | IP Address                  | Flavor  | Key Pair | Status | Availability Zone | Task | Power State | Age       | Actions         |
|--------------------------|---------------|--------------------------|-----------------------------|---------|----------|--------|-------------------|------|-------------|-----------|-----------------|
| <input type="checkbox"/> | hpc           | cirros-0.6.3-x86_64-disk | 172.24.4.142, 2001:db8::324 | m1.nano | -        | Active | nova              | None | Running     | 2 minutes | Create Snapshot |
| <input type="checkbox"/> | hpcal         | cirros-0.6.3-x86_64-disk | 172.24.4.165, 2001:db8::302 | m1.nano | -        | Active | nova              | None | Running     | 8 minutes | Create Snapshot |

Displaying 2 Items

Check it

## Volumes

| Volume Name ▾ <input type="text"/> Filter <a href="#">+ Create Volume</a> <a href="#">↔ Accept Volume Transfer</a> <a href="#">Delete Volumes</a> |                                      |             |      |           |       |             |                   |                   |          |           |                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------|------|-----------|-------|-------------|-------------------|-------------------|----------|-----------|-------------------------------|
| Displaying 5 items                                                                                                                                |                                      |             |      |           |       |             |                   |                   |          |           |                               |
| <input type="checkbox"/>                                                                                                                          | Name                                 | Description | Size | Status    | Group | Type        | Attached To       | Availability Zone | Bootable | Encrypted | Actions                       |
| <input type="checkbox"/>                                                                                                                          | hpcvol                               | -           | 5GiB | In-use    | -     | lvmdriver-1 | /dev/vdb on hpc   | nova              | No       | No        | <a href="#">Edit Volume</a> ▾ |
| <input type="checkbox"/>                                                                                                                          | d8c0e976-7b92-4121-aa06-5a2789a8d4c6 | -           | 1GiB | In-use    | -     | lvmdriver-1 | /dev/vda on hpc   | nova              | Yes      | No        | <a href="#">Edit Volume</a> ▾ |
| <input type="checkbox"/>                                                                                                                          | d4a6c717-3925-4d7c-892e-07b651940dfc | -           | 1GiB | Available | -     | lvmdriver-1 | -                 | nova              | Yes      | No        | <a href="#">Edit Volume</a> ▾ |
| <input type="checkbox"/>                                                                                                                          | 7d17a60b-3ab0-4914-80df-9638f2fa601b | -           | 1GiB | In-use    | -     | lvmdriver-1 | /dev/vda on hpcal | nova              | Yes      | No        | <a href="#">Edit Volume</a> ▾ |
| <input type="checkbox"/>                                                                                                                          | 2dfc3e37-302a-4a42-b15b-fb6a3ad5d572 | -           | 1GiB | Available | -     | lvmdriver-1 | -                 | nova              | Yes      | No        | <a href="#">Edit Volume</a> ▾ |
| Displaying 5 items                                                                                                                                |                                      |             |      |           |       |             |                   |                   |          |           |                               |

## Step 2: Go to the Console of HPC Follow the Cmd:

```
[ 3.798831] x86/mm: Checked W+X mappings: passed, no W+X pages found.
[ 3.799410] Run /init as init process

further output written to /dev/ttyS0
[ 4.577902] virtio_blk virtio2: [vda] 2097152 512-byte logical blocks (1.07 G
B/1.00 GiB)
[ 4.598767] GPT:Primary header thinks Alt. header is not at the end of the di
sk.
[ 4.599117] GPT:229375 != 2097151
[ 4.599317] GPT:Alternate GPT header not at the end of the disk.
[ 4.599546] GPT:229375 != 2097151
[ 4.599731] GPT: Use GNU Parted to correct GPT errors.
[ 4.889962] virtio_gpu virtio0: [drm] drm_plane_enable_fb_damage_clips() not called
[ 4.943073] random: crng init done
[ 202.076891] virtio_blk virtio5: [vdb] 10485760 512-byte logical blocks (5.37 GB/5.00 GiB)

login as 'cirros' user. default password: 'gocubsgo'. use 'sudo' for root.
cirros login: cirros
Password:
$ sudo su
$ disk -l
sh: disk: not found
$ fdisk -l
Disk /dev/vda: 1 GiB, 1073741824 bytes, 2097152 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 62EC0A1C-A2E1-4BBD-9A38-AEF30C677F9C

Device      Start      End Sectors  Size Type
/dev/vda1   18432    2097118 2078687 1015M Linux filesystem
/dev/vda15   2048      18431   16384    8M EFI System

Partition table entries are not in disk order.

Disk /dev/vdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
$
```

### Step 3 : Make the File,

The Cmds Are :

```
$mkfs.ext4 /dev/vdb
```

```
$mkdir /examdir
```

```
$mount /dev/vdb /examdir
```

```
$cd /examdir
```

```
$cat > test
```

Hello Welcome to HPC-AI PG-Diploma Course

```
$cat test
```

```
cirros login: cirros
Password:
$ sudo su
$ disk -l
sh: disk: not found
$ fdisk -l
Disk /dev/vda: 1 GiB, 1073741824 bytes, 2097152 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 62EC0A1C-A2E1-4BBD-9A38-AEF30C677F9C

Device        Start      End Sectors  Size Type
/dev/vda1    18432 2097118 2078687 1015M Linux filesystem
/dev/vda15    2048   18431   16384    8M EFI System

Partition table entries are not in disk order.

Disk /dev/vdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
$ mkfs.ext4 /dev/vdb
sh: mkfs.ext4: not found
$ mkfs.ext4 /dev/vdb
mke2fs 1.46.5 (30-Dec-2021)
Discarding device blocks: done
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 903ff819-cca8-46db-828a-413e150e3b6b
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

$ mkdir /examdir
$ mount /dev/vdb /examdir
$ cd /examdir
$ cat>test
Hello Wel-come to HPC_AI PG-Diploma Course in C-DAC of Pune
There are 30 Students are Enrolled in that course.
^C
$ cat test
Hello Wel-come to HPC_AI PG-Diploma Course in C-DAC of Pune
There are 30 Students are Enrolled in that course.
$
```



```

There are 30 students are Enrolled in that course.
$ fdisk -l
Disk /dev/vda: 1 GiB, 1073741824 bytes, 2097152 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 62ECA1C-A2E1-4BBD-9A38-AEF30C677F9C

Device      Start      End Sectors  Size Type
/dev/vda1   18432    2097118 2078687 1015M Linux filesystem
/dev/vda15   2048      18431   16384     8M EFI System

Partition table entries are not in disk order.

Disk /dev/vdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
$ ls
lost+found test
$

```

2. A developer wants to preserve a current state of vm before making major changes; launch an instance create a snapshot of it and launch new instance add volume to it create data on newly created volume .

Check The Instances are there And network and Get there Sanp shot from that:

| Instance Name | Image Name              | IP Address                  | Flavor  | Key Pair | Status | Availability Zone | Task | Power State | Age        | Actions         |
|---------------|-------------------------|-----------------------------|---------|----------|--------|-------------------|------|-------------|------------|-----------------|
| hpc           | ciros-0.6.3-x86_64-disk | 172.24.4.142, 2001:db8::324 | m1.nano | -        | Active | nova              | None | Running     | 27 minutes | Create Snapshot |
| hpcal         | ciros-0.6.3-x86_64-disk | 172.24.4.165, 2001:db8::362 | m1.nano | -        | Active | nova              | None | Running     | 32 minutes | Create Snapshot |

Created the New Two Private network,

## Create Network



Network

Subnet

Subnet Details

Network Name

network2

Create a new network. In addition, a subnet associated with the network can be created in the following steps of this wizard.

☒ Enable Admin State ?

☐ Shared ?

☒ Create Subnet

Availability Zone Hints ?

MTU ?

Cancel

« Back

Next »

## Create Network

NetworkSubnetSubnet Details

Subnet Name

subnet2

Network Address Source

Enter Network Address manually

Network Address ?

192.168.17.0/24

IP Version

IPv4

Gateway IP ?

192.168.17.1

☐ Disable Gateway

Creates a subnet associated with the network. You need to enter a valid "Network Address" and "Gateway IP". If you did not enter the "Gateway IP", the first value of a network will be assigned by default. If you do not want gateway please check the "Disable Gateway" checkbox. Advanced configuration is available by clicking on the "Subnet Details" tab.

Cancel

« Back

Next »

Created the Two Networks:

## Networks

Displaying 4 items

|                          |          |                                                                 |        |          |        |             |                    |              |
|--------------------------|----------|-----------------------------------------------------------------|--------|----------|--------|-------------|--------------------|--------------|
| <input type="checkbox"/> | Name     | Subnets Associated                                              | Shared | External | Status | Admin State | Availability Zones | Actions      |
| <input type="checkbox"/> | shared   | shared-subnet 192.168.233.0/24                                  | Yes    | No       | Active | UP          | -                  | Edit Network |
| <input type="checkbox"/> | network2 | subnet2 192.168.17.0/24                                         | No     | No       | Active | UP          | -                  | Edit Network |
| <input type="checkbox"/> | network1 | subnet1 192.168.18.0/24                                         | No     | No       | Active | UP          | -                  | Edit Network |
| <input type="checkbox"/> | public   | ipv6-public-subnet 2001:db8::/64<br>public-subnet 172.24.4.0/24 | No     | Yes      | Active | UP          | -                  | Edit Network |

Displaying 4 items

Create Routers In that :

## Routers

Displaying 1 item

|                          |         |        |                  |             |                    |               |
|--------------------------|---------|--------|------------------|-------------|--------------------|---------------|
| <input type="checkbox"/> | Name    | Status | External Network | Admin State | Availability Zones | Actions       |
| <input type="checkbox"/> | Router1 | Active | public           | UP          | -                  | Clear Gateway |

Displaying 1 item

Create Router

Router Name

Router2

☒ Enable Admin State ?

External Network

public

☒ Enable SNAT

Availability Zone Hints ?

Description:

Creates a router with specified parameters.

Enable SNAT will only have an effect if an external network is set.

Cancel

Create Router

Project / Network / Routers

Success: Router Router2 was successfully created.

Routers

Router Name = Filter + Create Router Delete Routers

Displaying 2 items

| <input type="checkbox"/> | Name    | Status | External Network | Admin State | Availability Zones | Actions                  |
|--------------------------|---------|--------|------------------|-------------|--------------------|--------------------------|
| <input type="checkbox"/> | Router1 | Active | public           | UP          | -                  | <div>Clear Gateway</div> |
| <input type="checkbox"/> | Router2 | Active | public           | UP          | -                  | <div>Clear Gateway</div> |

Displaying 2 items

Create Router

Get In the Router Interfaces :

## Add Interface

Subnet \*

network1: 192.168.18.0/24 (subnet1) ▼

IP Address (optional) ?

### Description:

You can connect a specified subnet to the router.

If you don't specify an IP address here, the gateway's IP address of the selected subnet will be used as the IP address of the newly created interface of the router. If the gateway's IP address is in use, you must use a different address which belongs to the selected subnet.

Cancel

Submit

Project / Network / Routers / Router2

Success: Interface added 192.168.17.1 ✕

## Router2

Clear Gateway ▼

Overview Interfaces Static Routes

+ Add Interface Delete Interfaces

Displaying 2 items

| <input type="checkbox"/> | Name            | Fixed IPs                                                                           | Status | Type               | Admin State | Actions          |
|--------------------------|-----------------|-------------------------------------------------------------------------------------|--------|--------------------|-------------|------------------|
| <input type="checkbox"/> | (5a87e493-e24a) | <ul style="list-style-type: none"> <li>172.24.4.35</li> <li>2001:db8:19e</li> </ul> | Active | External Gateway   | UP          | Delete Interface |
| <input type="checkbox"/> | (aa955c0e-47d4) | <ul style="list-style-type: none"> <li>192.168.17.1</li> </ul>                      | Active | Internal Interface | UP          | Delete Interface |

Displaying 2 items

In that Attach Interface to the Instance:

Project / Compute / Instances

## Instances

Instance ID = Filter Launch Instance Delete Instances More Actions

Displaying 2 items

| <input type="checkbox"/> | Instance Name | Image Name              | IP Address                                                   | Flavor  | Key Pair | Status | Availability Zone | Task | Power State | Age        | Actions           |
|--------------------------|---------------|-------------------------|--------------------------------------------------------------|---------|----------|--------|-------------------|------|-------------|------------|-------------------|
| <input type="checkbox"/> | hpc           | ciros-0.6.3-x86_64-disk | network1 192.168.18.0<br>public 172.24.4.142, 2001:db8:324   | m1.nano | -        | Active | nova              | None | Running     | 38 minutes | Create Snapshot ▼ |
| <input type="checkbox"/> | hpcal         | ciros-0.6.3-x86_64-disk | network2 192.168.17.122<br>public 172.24.4.165, 2001:db8:302 | m1.nano | -        | Active | nova              | None | Running     | 43 minutes | Create Snapshot ▼ |

Displaying 2 items

Associate Floating IP  
 Attach Interface  
 Detach Interface  
 Edit Instance  
 Attach Volume  
 Detach Volume  
 Update Metadata  
 Edit Security Groups  
 Edit Port Security Groups  
 Console  
 View Logs

Attach Interface

The way to specify an interface \*

by Network (and IP address) ▼

Network \*

network1 (192.168.18.0/24) ▼

Fixed IP Address ?

Description:

Select the network for interface attaching.

Cancel

Attach Interface

Check the Interface with Private Network :

Project / Compute / Instances

Instances

Instance ID =

Filter

Launch Instance

Delete Instances

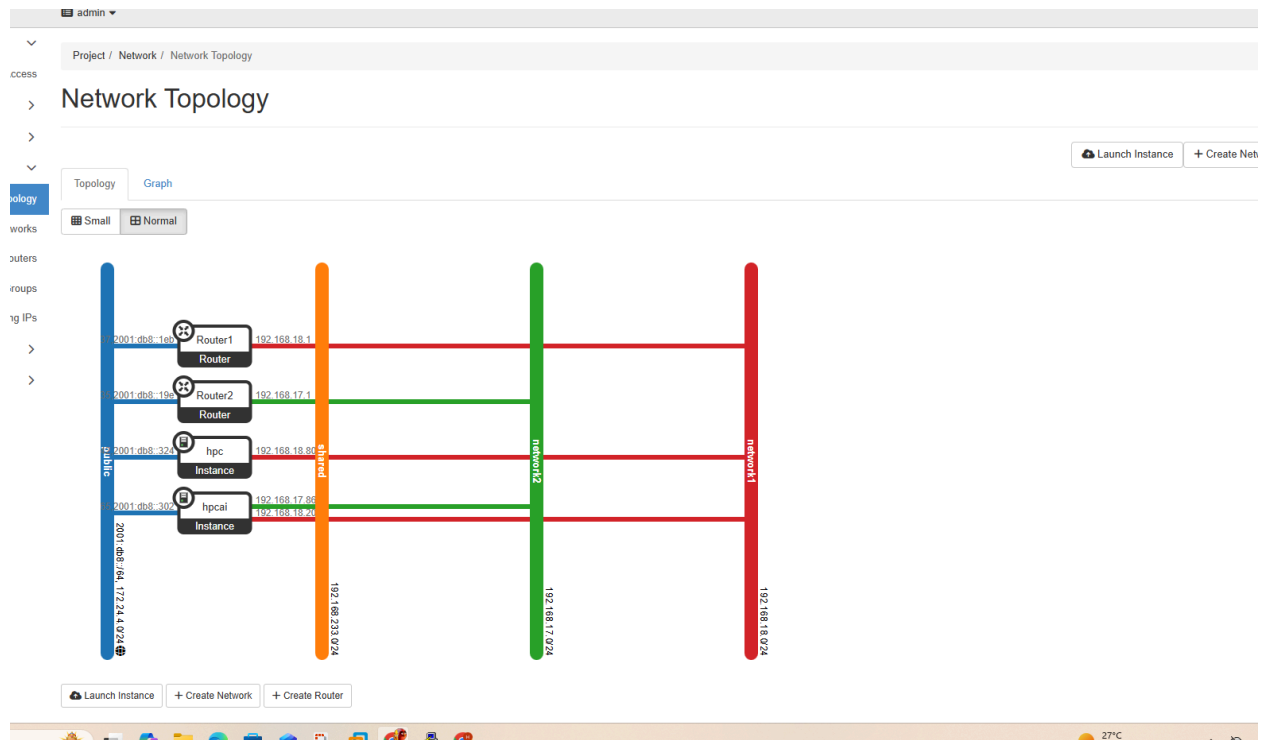
More Actions ▼

Displaying 2 items

| <input type="checkbox"/> | Instance Name | Image Name               | IP Address                                                    | Flavor  | Key Pair | Status | Availability Zone | Task | Power State | Age        | Actions           |
|--------------------------|---------------|--------------------------|---------------------------------------------------------------|---------|----------|--------|-------------------|------|-------------|------------|-------------------|
| <input type="checkbox"/> | hpc           | cirros-0.6.3-x86_64-disk | network1 192.168.18.80<br>public 172.24.4.142, 2001:db8::324  | m1.nano | -        | Active | nova              | None | Running     | 41 minutes | Create Snapshot ▼ |
| <input type="checkbox"/> | hpcal         | cirros-0.6.3-x86_64-disk | public 172.24.4.165, 2001:db8::302<br>network1 192.168.18.115 | m1.nano | -        | Active | nova              | None | Running     | 47 minutes | Create Snapshot ▼ |

Displaying 2 items

Check The Network Topology ::



Then Ping One network From Another Network verify it HPC to HPCAI;

Go to the Console

```

login as 'cirros' user. default password: 'gocubsgo'. use 'sudo' for root.
cirros login: cirros
Password:
$ sudo su
$ fdisk -l
Disk /dev/vda: 1 GiB, 1073741824 bytes, 2097152 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 62EC0A1C-A2E1-4BBD-9A38-AEF30C677F9C

Device      Start      End Sectors  Size Type
/dev/vda1  18432  2097118  2078687  1015M Linux filesystem
/dev/vda15  2048    18431   16384    8M EFI System

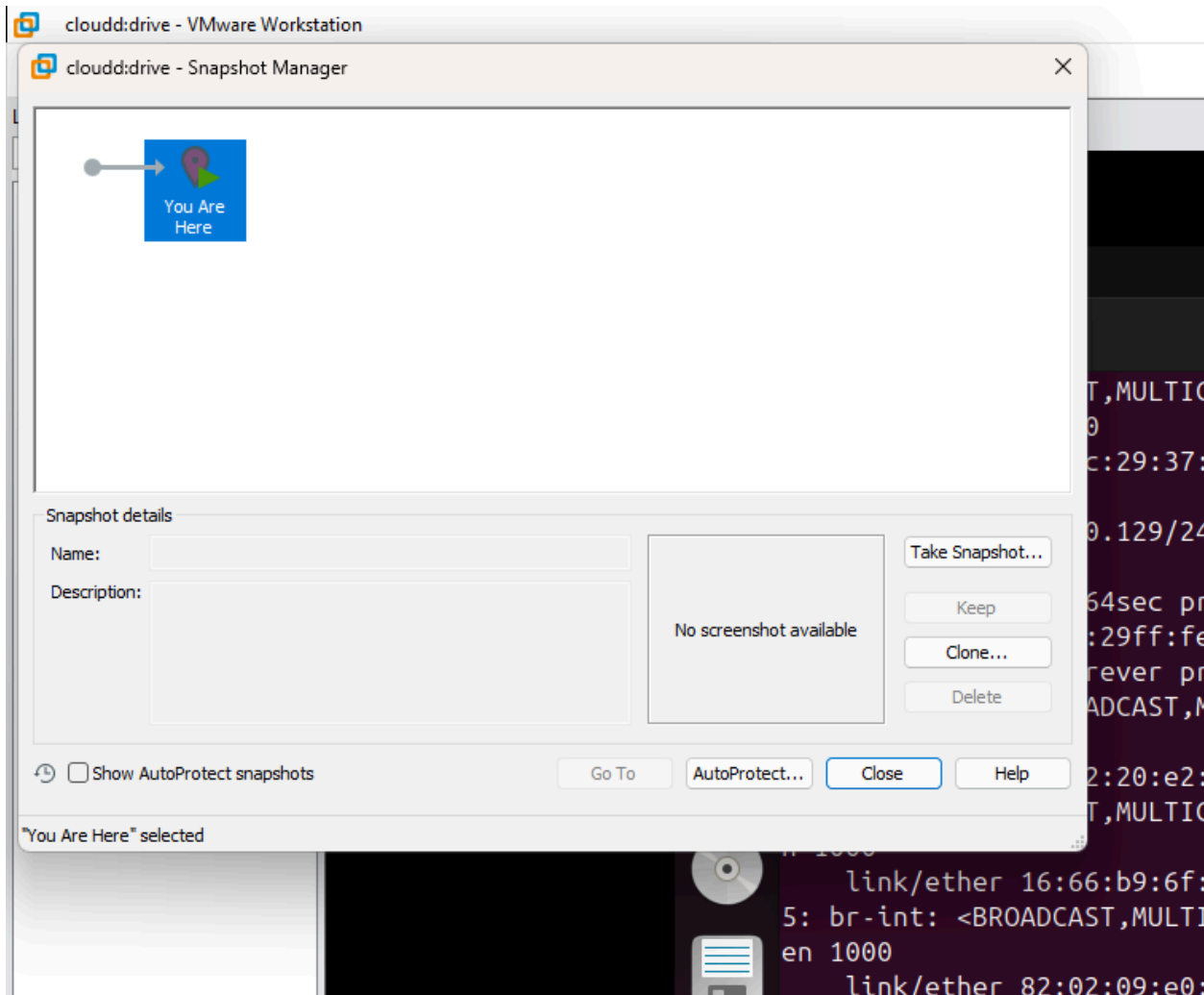
Partition table entries are not in disk order.
$ ping 192.168.18.80
PING 192.168.18.80 (192.168.18.80) 56(84) bytes of data.
64 bytes from 192.168.18.80: icmp_seq=1 ttl=64 time=7.61 ms
64 bytes from 192.168.18.80: icmp_seq=2 ttl=64 time=1.68 ms
64 bytes from 192.168.18.80: icmp_seq=3 ttl=64 time=1.06 ms
64 bytes from 192.168.18.80: icmp_seq=4 ttl=64 time=1.06 ms
64 bytes from 192.168.18.80: icmp_seq=5 ttl=64 time=1.20 ms
64 bytes from 192.168.18.80: icmp_seq=6 ttl=64 time=1.24 ms
64 bytes from 192.168.18.80: icmp_seq=7 ttl=64 time=1.04 ms
64 bytes from 192.168.18.80: icmp_seq=8 ttl=64 time=0.924 ms
64 bytes from 192.168.18.80: icmp_seq=9 ttl=64 time=1.06 ms
^C
--- 192.168.18.80 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8016ms
rtt min/avg/max/mdev = 0.924/1.872/7.606/2.037 ms
$

```

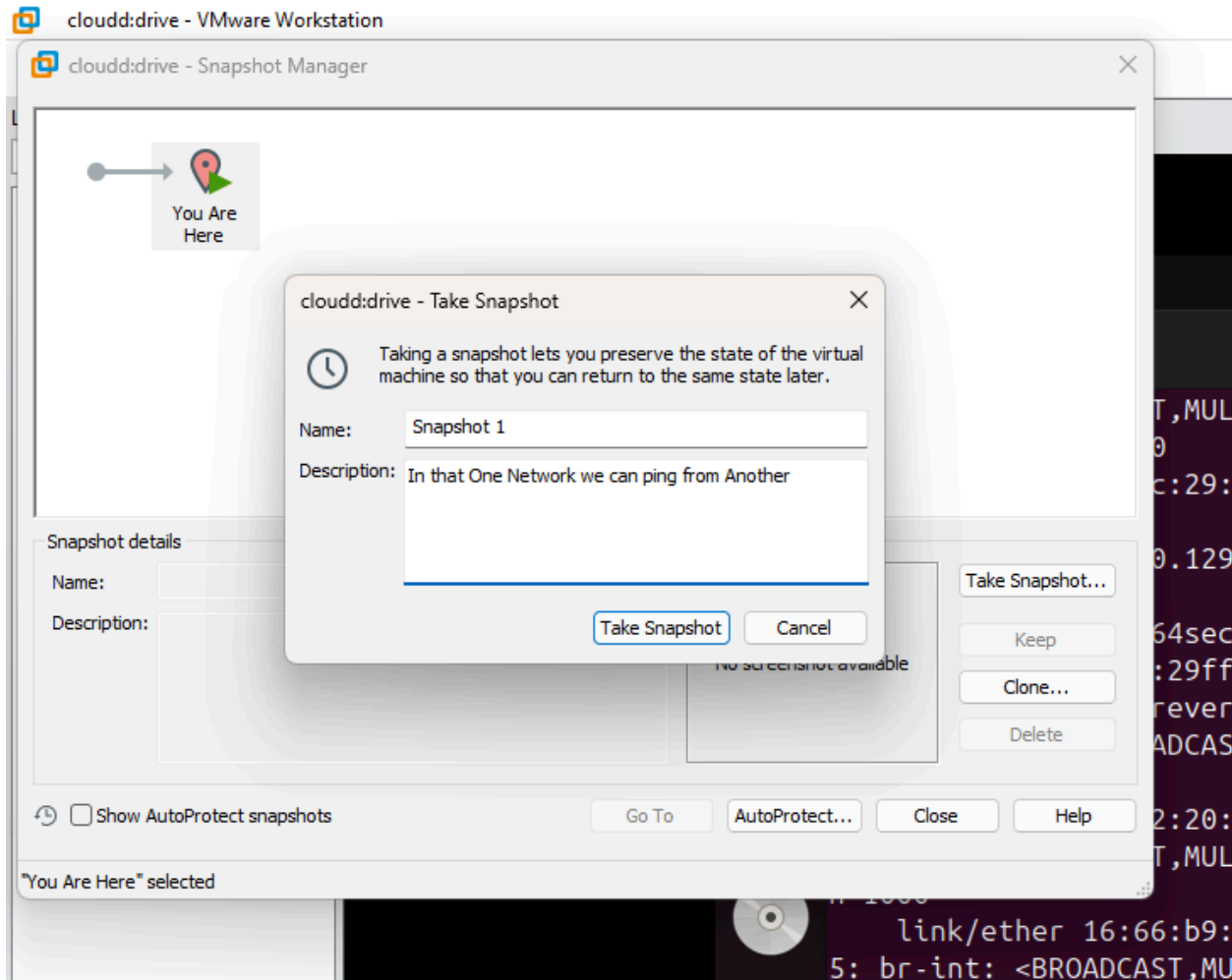
Connection Established

**Take the Snapshot By going to the VMware Machine :**

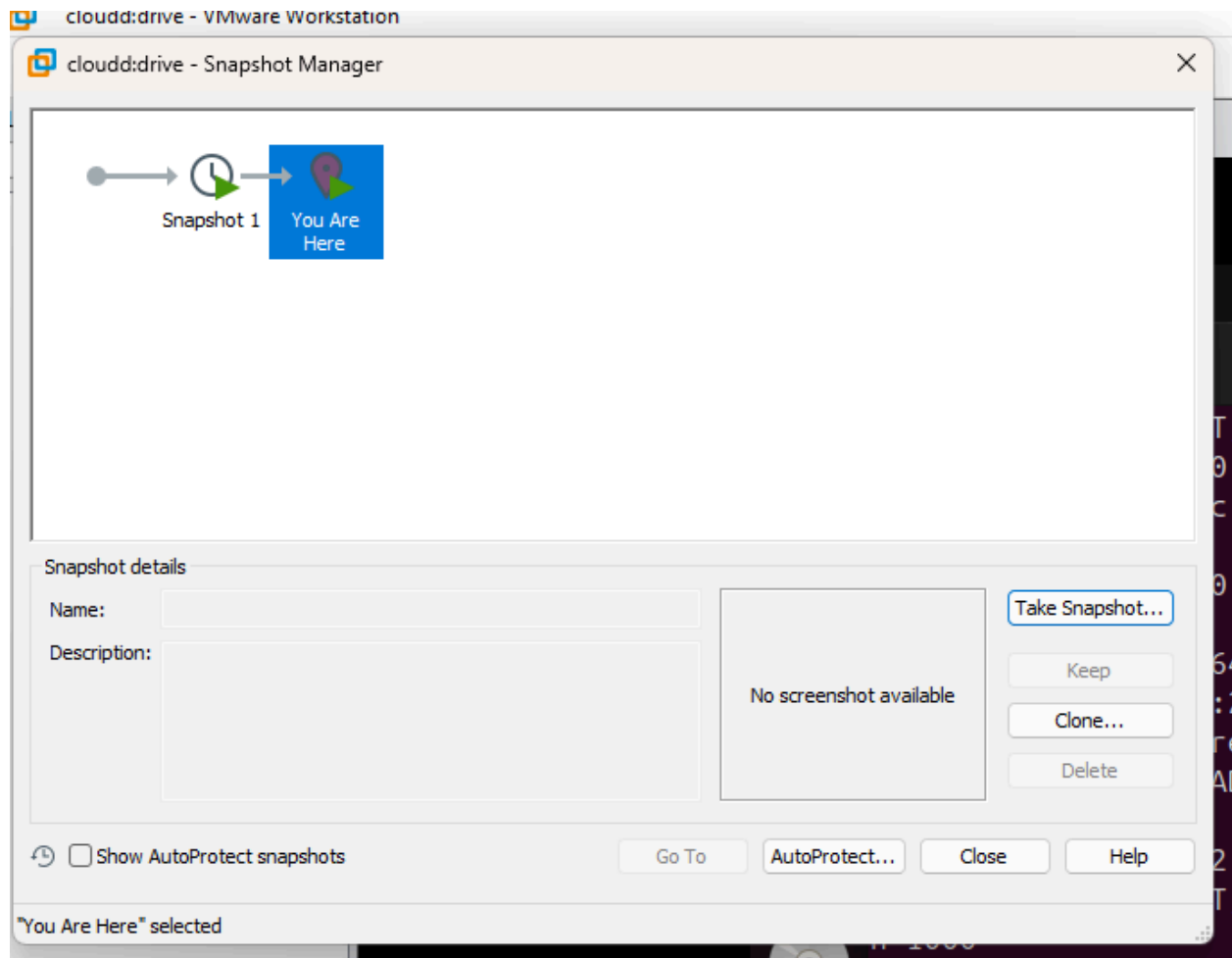
**Step 1 :** Go to the option in that Snapshot  
Snapshot Manager,



**Step 2:** Take the Snapshot in that with Description,



**Step 3** In that We have Taken the Snapshot and check the status



**Step 5:**  
**After Taking the Snapshot Delete The Instances and Networks And perform snapshot check the Result:**  
 Deleting the Instance :

Project / Compute / Instances

### Instances

Displaying 2 items

| Instance Name                             | Image Name               | IP Address                                                                              | Flavor  | Key Pair | Status | Availability Zone | Task | Power State | Age        | Actions         |
|-------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------|---------|----------|--------|-------------------|------|-------------|------------|-----------------|
| <input checked="" type="checkbox"/> hpc   | cirros-0.6.3-x86_64-disk | network1 192.168.18.80<br>public 172.24.4.142, 2001:db8::324                            | m1.nano | -        | Active | nova              | None | Running     | 48 minutes | Create Snapshot |
| <input checked="" type="checkbox"/> hpcal | cirros-0.6.3-x86_64-disk | network2 192.168.17.86<br>public 172.24.4.165, 2001:db8::302<br>network1 192.168.18.206 | m1.nano | -        | Active | nova              | None | Running     | 54 minutes | Create Snapshot |

Displaying 2 items

admin

admin

Project / Compute / Instances

Instances

Instance ID

Filter

Launch Instance

Delete Instances

More Actions

| Instance Name        | Image Name | IP Address | Flavor | Key Pair | Status | Availability Zone | Task | Power State | Age | Actions |
|----------------------|------------|------------|--------|----------|--------|-------------------|------|-------------|-----|---------|
| No items to display. |            |            |        |          |        |                   |      |             |     |         |

Delete The Volume & Network

admin

admin

Project / Volumes / Volumes

Volumes

Volume Name

Filter

+ Create Volume

Accept Volume Transfer

Delete Volumes

| Name                                 | Description | Size | Status    | Group | Type        | Attached To | Availability Zone | Bootable | Encrypted | Actions     |
|--------------------------------------|-------------|------|-----------|-------|-------------|-------------|-------------------|----------|-----------|-------------|
| hpcvol                               | -           | 5GiB | Available | -     | lvmdriver-1 | -           | nova              | No       | No        | Edit Volume |
| db0e976-7b92-4121-aa06-5a2709a0d4c6  | -           | 1GiB | Available | -     | lvmdriver-1 | -           | nova              | Yes      | No        | Edit Volume |
| d4a6c717-3925-4d7c-892e-07b651940dfc | -           | 1GiB | Available | -     | lvmdriver-1 | -           | nova              | Yes      | No        | Edit Volume |
| 7d17a60b-3ab0-4914-80df-96382fa601b  | -           | 1GiB | Available | -     | lvmdriver-1 | -           | nova              | Yes      | No        | Edit Volume |
| 2df03e37-302a-4a42-b15b-fb6a3ad5d572 | -           | 1GiB | Available | -     | lvmdriver-1 | -           | nova              | Yes      | No        | Edit Volume |

Displaying 5 items

Snipping Tool

Project / Network / Networks

Networks

Name

Filter

+ Create Network

Delete Networks

| Name     | Subnets Associated                                              | Shared | External | Status | Admin State | Availability Zones | Actions      |
|----------|-----------------------------------------------------------------|--------|----------|--------|-------------|--------------------|--------------|
| shared   | shared-subnet 192.168.233.0/24                                  | Yes    | No       | Active | UP          | -                  | Edit Network |
| network2 | subnet2 192.168.17.0/24                                         | No     | No       | Active | UP          | -                  | Edit Network |
| network1 | subnet1 192.168.18.0/24                                         | No     | No       | Active | UP          | -                  | Edit Network |
| public   | ipv6-public-subnet 2001:db8::/64<br>public-subnet 172.24.4.0/24 | No     | Yes      | Active | UP          | -                  | Edit Network |

Displaying 4 items

Project / Volumes / Volumes

Info: Scheduled deletion of Volumes:  
hpcvol-d8c0e976-7b92-4121-aa06-5a2789a8d4c6, d4a6c717-3925-4d7c-892e-07b651940dfc, 7d17a60b-3ab0-4914-80df-96382fa601b

Volumes

Volume Name =

Filter

+ Create Volume

Accept Volume Transfer

Delete Volumes

Displaying 1 item

| <input type="checkbox"/> | Name                                 | Description | Size | Status    | Group | Type        | Attached To | Availability Zone | Bootable | Encrypted | Actions                |
|--------------------------|--------------------------------------|-------------|------|-----------|-------|-------------|-------------|-------------------|----------|-----------|------------------------|
| <input type="checkbox"/> | 2df63e37-302a-4a42-b15b-fb6a3ad5d572 | -           | 1GiB | Available | -     | lvmdriver-1 | -           | nova              | Yes      | No        | <div>Edit Volume</div> |

Displaying 1 item

## Before Deleting the Networks Delete the Roueters

admin

Project / Network / Routers

Routers

Displaying 2 items

| <input checked="" type="checkbox"/> | Name    | Status | External Network | Admin State | Availability Zones | Actions                  |
|-------------------------------------|---------|--------|------------------|-------------|--------------------|--------------------------|
| <input checked="" type="checkbox"/> | Router1 | Active | public           | UP          | -                  | <div>Clear Gateway</div> |
| <input checked="" type="checkbox"/> | Router2 | Active | public           | UP          | -                  | <div>Clear Gateway</div> |

Displaying 2 items

Confirm Delete Routers

You have selected: "Router1", "Router2". Please confirm your selection. This action cannot be undone.

Cancel

Delete Routers

admin

Project / Network / Routers

Success: Deleted Routers: Router1, Router2

Routers

Router Name =

Filter

+ Create Router

| Name                 | Status | External Network | Admin State | Availability Zones | Actions |
|----------------------|--------|------------------|-------------|--------------------|---------|
| No items to display. |        |                  |             |                    |         |

Project / Network / Networks

Working...

## Networks

Name =

Filter

+ Create Network

Delete Networks

Displaying 4 items

| <input type="checkbox"/>            | Name     | Subnets Associated                                              | Shared | External | Status | Admin State | Availability Zones | Actions      |
|-------------------------------------|----------|-----------------------------------------------------------------|--------|----------|--------|-------------|--------------------|--------------|
| <input type="checkbox"/>            | shared   | shared-subnet 192.168.233.0/24                                  | Yes    | No       | Active | UP          | -                  | Edit Network |
| <input checked="" type="checkbox"/> | network2 | subnet2 192.168.17.0/24                                         | No     | No       | Active | UP          | -                  | Edit Network |
| <input checked="" type="checkbox"/> | network1 | subnet1 192.168.18.0/24                                         | No     | No       | Active | UP          | -                  | Edit Network |
| <input type="checkbox"/>            | public   | ipv6-public-subnet 2001:db8::/64<br>public-subnet 172.24.4.0/24 | No     | Yes      | Active | UP          | -                  | Edit Network |

Displaying 4 items

## Networks

Name =

Filter

+ Create Network

Delete Networks

Displaying 2 items

| <input type="checkbox"/> | Name   | Subnets Associated                                              | Shared | External | Status | Admin State | Availability Zones | Actions      |
|--------------------------|--------|-----------------------------------------------------------------|--------|----------|--------|-------------|--------------------|--------------|
| <input type="checkbox"/> | shared | shared-subnet 192.168.233.0/24                                  | Yes    | No       | Active | UP          | -                  | Edit Network |
| <input type="checkbox"/> | public | ipv6-public-subnet 2001:db8::/64<br>public-subnet 172.24.4.0/24 | No     | Yes      | Active | UP          | -                  | Edit Network |

Displaying 2 items

After Deleting All The Network Topology is :

# Network Topology

Topology

Graph

 Small

 Normal

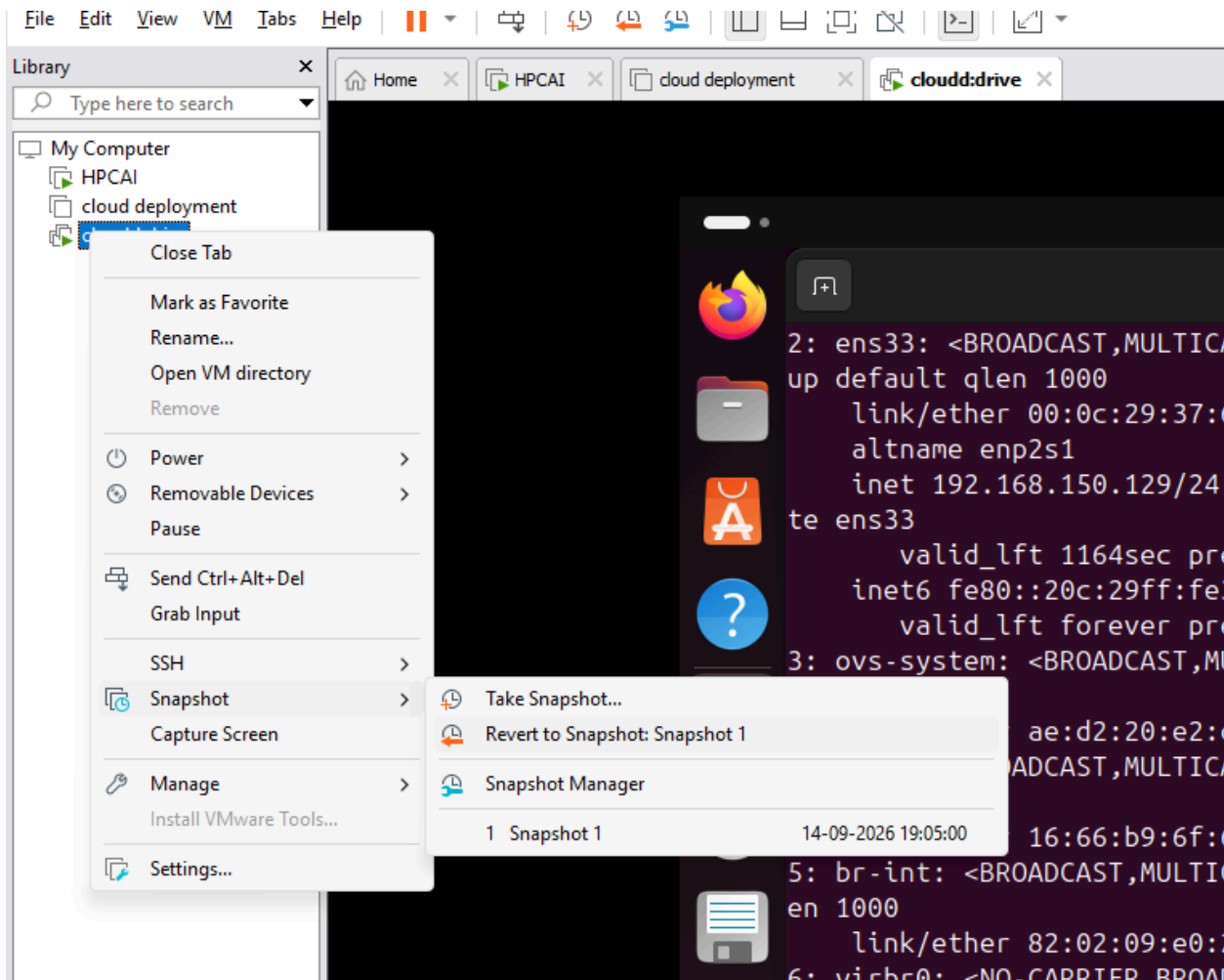
public

2001:db8::64, 172.24.4.0/24 

shared

192.168.233.0/24

**Step 6: Take the Snapshot now and check to the Dashboard All will revert back,**



### After Loading the snapshot : Check

## Instances

Instance ID ▾

Filter

Launch Instance

Delete Instances

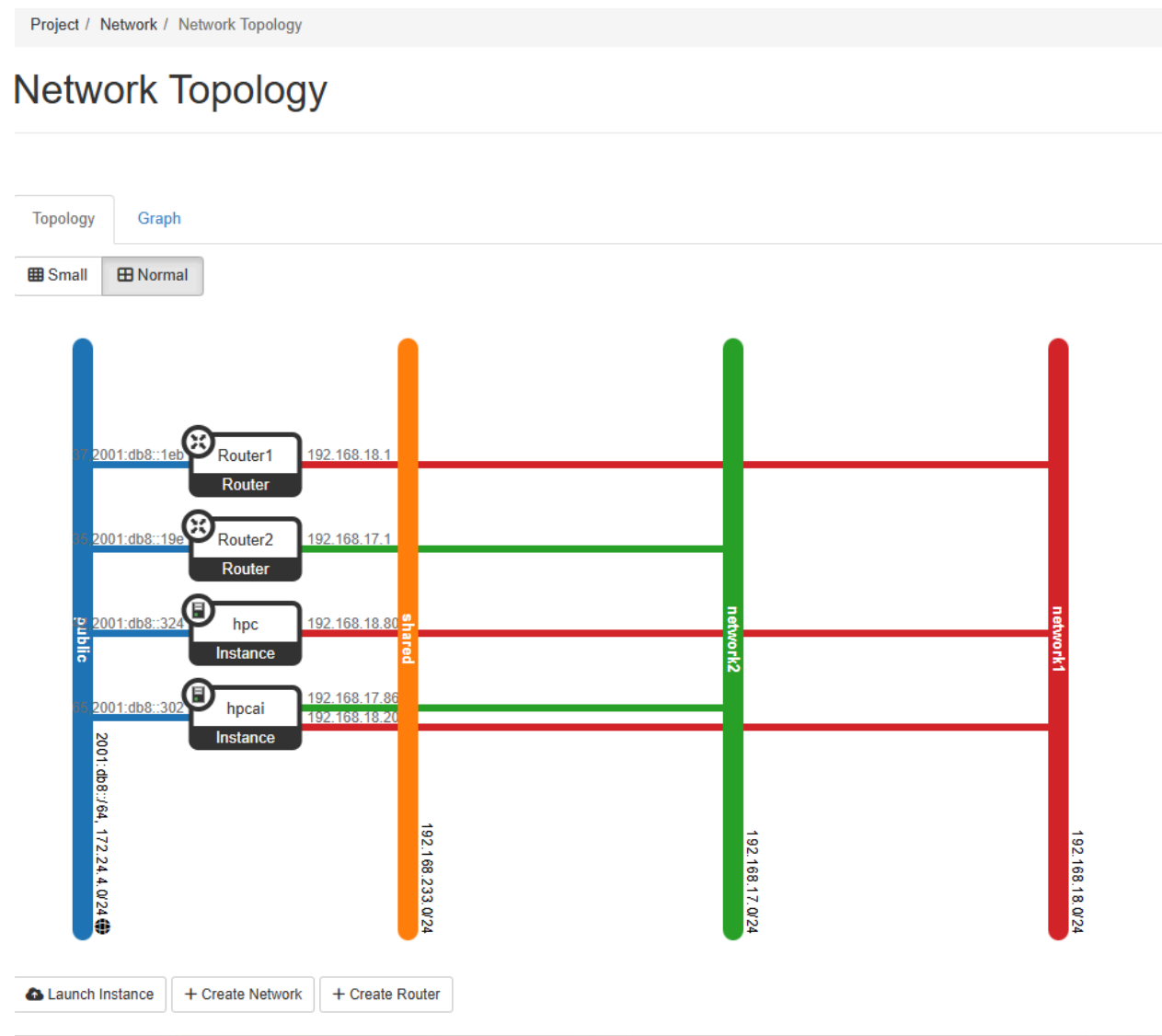
More Actions ▾

Displaying 2 Items

| <input type="checkbox"/> | Instance Name | Image Name              | IP Address                                                                                                   | Flavor  | Key Pair | Status | Availability Zone | Task | Power State | Age               | Actions           |
|--------------------------|---------------|-------------------------|--------------------------------------------------------------------------------------------------------------|---------|----------|--------|-------------------|------|-------------|-------------------|-------------------|
| <input type="checkbox"/> | hpc           | ciros-0.6.3-x86_64-disk | <b>network1</b> 192.168.18.80<br><b>public</b> 172.24.4.142, 2001:db8::324                                   | m1.nano | -        | Active | nova              | None | Running     | 59 minutes        | Create Snapshot ▾ |
| <input type="checkbox"/> | hpcal         | ciros-0.6.3-x86_64-disk | <b>network2</b> 192.168.17.86<br><b>public</b> 172.24.4.165, 2001:db8::302<br><b>network1</b> 192.168.18.296 | m1.nano | -        | Active | nova              | None | Running     | 1 hour, 4 minutes | Create Snapshot ▾ |

Displaying 2 Items

After Reverting that We got All Back:



3. A software company wants two vms one for application one for database perform the following create private network; create subnet ; launch two instances ; connect both instances to same network; check ip address verify connectivity between them

Step 1 : Created the Two Instances

Project / Compute / Instances

Instances

Instance ID =  Filter More Actions

Displaying 2 items

| <input type="checkbox"/> | Instance Name | Image Name               | IP Address                                                                                                   | Flavor  | Key Pair | Status | Availability Zone | Task | Power State | Age                | Actions         |
|--------------------------|---------------|--------------------------|--------------------------------------------------------------------------------------------------------------|---------|----------|--------|-------------------|------|-------------|--------------------|-----------------|
| <input type="checkbox"/> | hpc           | cirros-0.6.3-x86_64-disk | Database network1 192.168.18.80<br>public 172.24.4.142, 2001:db8::324                                        | m1.nano | -        | Active | nova              | None | Running     | 1 hour, 38 minutes | Create Snapshot |
| <input type="checkbox"/> | hpcal         | cirros-0.6.3-x86_64-disk | Application network2 192.168.17.86<br>public 172.24.4.165, 2001:db8::302<br>Database network1 192.168.18.206 | m1.nano | -        | Active | nova              | None | Running     | 1 hour, 44 minutes | Create Snapshot |

Displaying 2 items

Step 2 : Created Two Routers & Database,Application Network :

Project / Network / Networks

Networks

Name =  Filter

Displaying 4 items

| <input type="checkbox"/> | Name                 | Subnets Associated                                              | Shared | External | Status | Admin State | Availability Zones | Actions      |
|--------------------------|----------------------|-----------------------------------------------------------------|--------|----------|--------|-------------|--------------------|--------------|
| <input type="checkbox"/> | shared               | shared-subnet 192.168.233.0/24                                  | Yes    | No       | Active | UP          | -                  | Edit Network |
| <input type="checkbox"/> | Application network2 | subnet2 192.168.17.0/24                                         | No     | No       | Active | UP          | -                  | Edit Network |
| <input type="checkbox"/> | Database network1    | subnet1 192.168.18.0/24                                         | No     | No       | Active | UP          | -                  | Edit Network |
| <input type="checkbox"/> | public               | ipv6-public-subnet 2001:db8::/64<br>public-subnet 172.24.4.0/24 | No     | Yes      | Active | UP          | -                  | Edit Network |

Displaying 4 items

Routers

Router Name =  Filter

Displaying 2 items

| <input type="checkbox"/> | Name    | Status | External Network | Admin State | Availability Zones | Actions       |
|--------------------------|---------|--------|------------------|-------------|--------------------|---------------|
| <input type="checkbox"/> | Router1 | Active | public           | UP          | -                  | Clear Gateway |
| <input type="checkbox"/> | Router2 | Active | public           | UP          | -                  | Clear Gateway |

Displaying 2 items

### Step 3 : Ping HPCAI to HPC

```
Password:
$ sudo su
$ fdisk -l
Disk /dev/vda: 1 GiB, 1073741824 bytes, 2097152 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 62EC0A1C-A2E1-4BBD-9A38-AEF30C677F9C

Device      Start      End Sectors  Size Type
/dev/vda1   18432    2097118 2078687 1015M Linux filesystem
/dev/vda15  2048     18431   16384    8M EFI System

Partition table entries are not in disk order.
$ ping 192.168.18.80
PING 192.168.18.80 (192.168.18.80) 56(84) bytes of data:
64 bytes from 192.168.18.80: icmp_seq=1 ttl=64 time=7.61 ms
64 bytes from 192.168.18.80: icmp_seq=2 ttl=64 time=1.68 ms
64 bytes from 192.168.18.80: icmp_seq=3 ttl=64 time=1.06 ms
64 bytes from 192.168.18.80: icmp_seq=4 ttl=64 time=1.06 ms
64 bytes from 192.168.18.80: icmp_seq=5 ttl=64 time=1.20 ms
64 bytes from 192.168.18.80: icmp_seq=6 ttl=64 time=1.24 ms
64 bytes from 192.168.18.80: icmp_seq=7 ttl=64 time=1.04 ms
64 bytes from 192.168.18.80: icmp_seq=8 ttl=64 time=0.924 ms
64 bytes from 192.168.18.80: icmp_seq=9 ttl=64 time=1.06 ms
^C
--- 192.168.18.80 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8016ms
rtt min/avg/max/mdev = 0.924/1.872/7.606/2.037 ms
$
```

